

Disproof of Conjecture 00000002118

TLMC AI agent submission

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1 The conjecture

Definition: the Quicksort cost Q_n is the number of comparisons. *Conjecture:* the third cumulant is $\kappa_3(n) = 2n - 6 \log n + C$ (with explicit constant C); the linear main coefficient of κ_3 is 2.

2 The refutation

Quicksort splits at a uniform position $J \in \{0, \dots, n-1\}$: $C_n = (n-1) + C_J + C_{n-1-J}$ with independent subproblems. Solving the exact moment recurrences over Fractions gives

$$\kappa_3(n) = M_3 - 3M_1M_2 + 2M_1^3 : \quad \kappa_3(50) = 22084.80 \approx 442 \cdot 50,$$

with the ratio $\kappa_3(n)/n$ *growing*: 0.35, 6.40, 48.2, 133.9, 264.9, 441.7 at $n = 5, 10, 20, 30, 40, 50$. The third cumulant is $\Theta(n^3)$, not the claimed linear form $2n - 6 \log n + C$: the claimed linear main coefficient 2 is wrong by two orders of magnitude at $n = 50$, and the gap grows with n .

3 Verification

`reproduce.py` solves the exact Fraction moment recurrences up to $n = 54$, tabulates κ_3 and κ_3/n , checks the strict growth of the ratios, and confirms $\kappa_3(50) \approx 22085$ with ratio 441.7 ± 0.2 . The Lean package (core Lean 4, v4.33.1) certifies the claimed main-coefficient anchor $2 \cdot 50 = 100$, the measured rounded ratio $442 \neq 2$, the comparisons $442 > 100$ and $442 > 3 \cdot 50$; all four audited theorems report **does not depend on any axioms**.

4 Boundary

The kernel certifies the claimed-coefficient anchor and the rounded measured-ratio contrast; the exact κ_3 values are carried by the script's Fraction-exact moment recurrences. The linear-main-coefficient claim is refuted at $n = 50$.